MILITARY SPECIFICATION

Reproducer-Set, Voice Warning AN/ASH-19(*)

This specification has been approved
by the Bureau of Naval Weapons,
Department of the Navy.

## 1. SCOPE

1.1 Scope - The equipment covered by this specification shall accept signals from the aircraft's fault sensors and transmit human voice warning messages to the pilot through his headset, alerting him to hazardous or potentially hazardous conditions in the aircraft. The system provides twenty appropriate pre-recorded messages having a relative order of importance with every other message.

1.2 Classification - The Voice Warning Reproducer Set shall consist of the following items:

| Items | Type Designation | Appl. Para. |
|---|---|---|
| Reproducer-Converter, Voice Signal | RP-139(*)/ASH-19 | 3.4.1 |
| Mounting | MT-3290(*)/ASH-19 | 3.4.3 |

## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

| | |
|---|---|
| MIL-E-4682 | Electron Tubes and Transistors, Choice and Application of |
| MIL-W-5088 | Wiring; Aircraft, Installation of |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-M-7793 | Meter, Time Totalizing |



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|---|---|
| MIL-C-172 | **Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft.)** |
| MIL-E-17555 | Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of |
| MIL-T-18303 | Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment |
| MIL-R-23094 | Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for |
| MIL-S-23603 | System Readiness/Maintainability; Avionics System Design, General specification for |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-704 | Electric Power, Aircraft Characteristics and Utilization of |
| MS-17321 | Meter, Time Totalizing, 28VDC, Miniature Digital |

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDI, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(3) When previously produced models of this equipment did



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not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 14.0 pounds.

3.2.2 Reliability -

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 400 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.2.3 Reliability In Mean Time Between Failures (MTBF) - For the test program under 4.4.3, the mean (operating) time between failures shall be 400 hours. Procedure II of Specification MIL-R-23094 shall be used.

3.2.2.4 Time Totalizing Meter - The equipment shall contain a time totalizing meter in accordance with Specification MIL-M-7793. The meter shall be of the type MS17321-2.

3.2.3 Cabling and Connections -

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.



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3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient ($25^{\circ}C \pm 5^{\circ}C$) |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90% relative humidity |

Input power voltage 27.5 $\pm$0.5 V DC

3.2.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment, except as modified herein:

3.2.9.1 Temperature - The equipment shall operate satisfactorily when subjected to a maximum temperature of +65°C. The storage temperature shall not exceed +65°C.

3.2.9.2 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and IV of Specification MIL-E-5400.



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3.2.10 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amount.

(1) D.C. Power 28V, Category "B" 1.75 Amps.

3.2.10.1 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.10.2 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3 Performance -

3.3.1 Operation - The Voice Warning Reproducer Set shall accept positive, direct current voltage signals from 50 aircraft fault sensors without deterioration to the normal aircraft fault warning system. It shall provide for the selection and play out of any one of twenty prerecorded voice warning messages, fifteen seconds in length, through audio outputs connected to the aircraft's intercommunication system. Each message channel shall have a predetermined priority sequence with every other channel and a higher priority message shall take over and play out over any message of lower priority. The message when activiated by the sensor shall continue playing until:

a. The activating fault is corrected.

b. A channel of higher priority is activated.

c. The message is silenced by an override switch.

d. The assembly is rearmed by the reset switch.

3.4 Detail Requirements -

3.4.1 Reproducer-Converter, Voice Signal RP-139/ASH-19 - The Reproducer-Converter shall meet the following requirements:

3.4.1.1 Function - The equipment shall accept voltages from the fault sensors and translate the fault to an audio message.

3.4.1.2 Form Factor - The equipment shall be enclosed in a case having the maximum dimensions of 9.75 inches long by 7.30 inches wide by 5.50 inches high or contained in a volume of 392 cu. in. maximum.

3.4.1.3 Weight - The weight of the equipment less mounting shall not exceed twelve pounds



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3.4.1.4 Electrical Connections - Connections to external circuits shall be provided by connectors meeting the requirements of Specification MIL-E-5400. The use of a minimum of two connectors is desirable, the function of one connector to connect all fault sensor inputs to the equipment and the function of the other to provide power input, audio output and outlet to control switches.

3.4.1.5 Inputs - There shall be provisions for fifty separate input signals from the fault sensors. The current drain for any one signal input shall not exceed 50 milliamperes at 28 volts D.C.

3.4.1.5.1 Input Signals - The nominal signal is a step change in voltage from zero to a plus 18-30 volts D.C. It must be recognized, however, that transducers are subject to modulation by variations in the signal source. Some signals exhibit slow rise characteristics and some switches have pronounced bounce. The input circuits shall accept as varied types of signals as practicable. The equipment shall not respond to an input signal of less than 5 volts or to any signal of less than 10 milliseconds duration.

3.4.1.5.2 Input Circuits - The input circuits shall be such that their common connection to existing signal sources shall not adversely affect existing visual warning system operation.

3.4.1.5.3 Grouping - Utilization of the fifty inputs and their grouping for connection to the twenty message channels will be determined by the procuring activity.

3.4.1.5.4 Priority - The equipment shall provide a predetermined priority of activation for each of the twenty message channels. Priority shall be from the lowest number channel to the highest number channel (1 to 20).

3.4.1.5.5 Input Changes - Provision shall be made for revising the connection and grouping of the input signals and for changing priority of warning messages. This shall be done without the necessity of changing aircraft wiring and with minimum change to the Voice Signal Reproducer-Converter. Maximum flexibility of connection shall be provided consistent with size and weight limitations.

3.4.1.6 Message Storage and Recovery - The equipment shall include a tape message storage and recovery unit with a capacity of twenty messages of 15 seconds minimum duration. Any activated message shall be played, from the beginning, as an audio signal in the output circuit.

3.4.1.6.1 Priority - A signal for a higher priority message shall cause discontinuance of the message in progress, and play the new message from the beginning. Circuitry shall be provided which will permit the combining of two or more lower priority similar warning messages and shall cause a message of higher priority to be played.



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3.4.1.6.2 Delay - There shall be no more than 0.5 second delay between any command signal and appearance of its associated output audio signal voltage.

3.4.1.6.3 Erasure or Recording - The equipment shall include no provision for erasure or recording.

3.4.1.6.4 Tape Replacement - The message storage tape shall be replaceable with minimum disassembly and checking of the Voice Signal Reproducer-Converter.

3.4.1.6.5 Self Test Provision - The equipment shall be capable of a complete system checkout without any faults being present. Each warning message shall be played in the order of its priority.

3.4.1.7 Audio Output - The voice message shall be fed to the outputs as an amplified audio frequency voltage. There shall be three isolated outputs each of which shall have the following characteristics:

Rated Power - 250 milliwatts into a 600-ohm load at 1000 cps.

Frequency range - 400 cycles to 3000 cycles.

The outputs shall be two terminal, ungrounded type. Each output shall present an open circuit when no message is in play. In addition, the design shall preclude feedback of D.C. into the interphone system and shall provide for compatibility with the intercommunication systems in naval aircraft such as AN/AIC-14 and AN/AIC-22(V).

3.4.1.7.1 Volume Adjustment - Provision shall be made for volume adjustment of the audio output. The volume control shall be screw-driver type, located on the front face of the equipment, and shall have locking provisions. It shall provide 12 ±3 db attenuation of the audio output power and shall provide essentially linear attenuation with shaft rotation. Appropriate markings for adjustment shall be provided.

3.4.1.7.2 Audio Quality - Voice messages shall have good intelligibility when played under any of the environmental conditions of paragraph 3.2.9. The characteristics of the audio output under standard conditions shall be:

(a) Power Level - At any output frequency within the specified range, the power output shall be within ±6 db of that obtained at 1000 cycles per second (cps). The response below 400 cps and above 3000 cps shall be attenuated as much as practical without the use of large filter components.



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(b) Signal to Noise Ratio - 30 db minimum at any setting of the output volume adjustment.

(c) Stability - Output shall not vary from setting by more than ±3 db.

(d) Flutter - The flutter content shall not be greater than 3.0 per cent. RMS

(e) WOW - The WOW content shall not be greater than 3.0 per cent. RMS

(f) Distortion - Shall not exceed 10 per cent at any setting of the volume adjustment.

3.4.1.7.3 Output Isolation - The equipment shall include switching contacts which shall operate to isolate the output amplifier from the intercommunication system except when a message is in play.

3.4.2 Controls - The equipment shall be controlled by the following external switches not supplied as part of this equipment:

(a) Power ON-OFF Switch - Single pole, single throw type. Controls operating power to the equipment.

(b) Override Switch - Momentary, normally open, single pole type. Override the channel in play.

(c) Reset Switch - Momentary, Normally closed, single pole type. Interrupts any message in play and reactivates all overridden channels on whcih fault signals are present.

(d) Test Switch - Momentary, normally open, single pole type. Energizes the test relay.

3.4.3 Mounting, MT-3290/ASH-19 - The Voice Signal Reproducer-Converter shall be capable of installation in any attitude in a mounting base dimensioned to retain the equipment. A multi-functional barrier vibration isolator of a flexible plastic type shall be employed subject to approval of the Bureau of Naval Weapons. The mounting base shall conform to the applicable requirements of MIL-C-172. Means shall be provided to mount or remove the Voice Signal Reproducer-Converter from the mounting base without special tools. The maximum weight of the mounting base shall be 2 pounds not including mating connectors.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Production Sample Tests
(3) Acceptance Tests
(4) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surfact temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.



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4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered at the contractors expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractors expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.



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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. Tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test



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4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

| | |
|---|---|
| Temperature | Ambient room |
| Humidity | Ambient room |
| Vibration | Any selected frequency within the range of 20 to 30 cps (excluding resonant points ) and a minimum amplitude of ± 3 g's |

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satsifactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action).



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| Quantity of Equipments Offered for Acceptance | Quantity to be Selected For Sampling Test |
|---|---|
| First 10 | 1 |
| Next 50 | 1 |
| Next 75 | 1 |
| Next 100 | 1 |
| | 1 for each additional 200 or fraction thereof |

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational Test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure II from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be III from Table I of Specification MIL-R-23094 except that vibration shall be limited to 20 cps and the length of the heating period shall be based on a stabilized high temperature of two hours. The duty cycle shall not exceed 4 hours.

4.4.3.3 Accept-Reject Criteria - Figure 11 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.



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4.4.3.4 Performance Characteristics to be Measured -

| Characteristic | Spec. Requirement | Time in cycle that measurement is to be made |
|---|---|---|
| Message Storage & recovery | 3.4.1.6 | Every fifteen minutes |
| Priority | 3.4.1.6.1 | Every fifteen minutes |
| Audio Quality | 3.4.1.7.2(b)<br>3.4.1.7.2(f) | At the beginning and end of each duty cycle. |

4.4.3.5 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests or Reliability Assurance Tests.

4.4.4.1 Special Test Schedule - Selection of equipment for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity, Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:



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(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

| Quantity of Equipments Offered for Acceptance | Quantity to be Selected For Life Test |
|---|---|
| First 25 | 1 |
| Next 175 | 1 |
| Next 300 | 1 |
| | 1 for each additional 500 or fraction thereof |

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

| | |
|---|---|
| Temperature | Normal room |
| Altitude | Normal ground (0 - 5000 ft.) |
| Humidity | Room ambient |
| **D. C. Voltage** | **27.0 ± 2.0 volts** |



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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. **The "Wear" items shall be determined by agreement between the contractor and the procuring activity.**

(2) **After reworking, the contractor shall resubmit the equipment for acceptance.**



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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contractor order in accordance with Specification MIL-E-17555.

## 6. NOTES

6.1 Intended Use - It is intended that this voice Warning Signal Assembly covert warning signal voltages from sensor elements in the aircraft electrical system to previously recorded warning messages which are played out through the aircraft's intercommunication system or to designated stations such as the pilot, copilot, etc.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 $\pm$ 0.5 V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each



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request shall be accompanied by complete supporting information. Such information shall include consideration of System Readiness/Maintainability requirements of MIL-S-23603(WEP).

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, may be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.7 The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:



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(a) Title, number, and date of this specification.

(b) Quantity of each of the items listed in 1.2 by type designation.

(c) Selection of applicable levels of packaging and packing.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-81000(Wep) Reproducer-Set, Vocie Warning AN/ASH-19 (*)

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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